

1) Covariance Analysis

Given Data:

 $x_i: 3.2, 4.0, 5.5, 4.8$ $y_i: 15, 18, 25, 20$

$$\text{Mean of } x = (3.2 + 4.0 + 5.5 + 4.8) / 4 = 4.375$$

$$\text{Mean of } y = (15 + 18 + 25 + 20) / 4 = 19.5$$

Calculation:

x_i	y_i	$x_i - \bar{X}$	$y_i - \bar{Y}$	$(x_i - \bar{X})(y_i - \bar{Y})$
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.42	0.5	0.2125

$$\sum (x_i - \bar{X})(y_i - \bar{Y}) = 12.25$$

$$\text{Covariance} = 12.25 / 4 = 3.0625$$

Answer:

Mean of $x = 4.375$

Mean of $y = 19.5$

Covariance = 3.0625

2) Data Smoothing using Equal-Frequency Binning

Given Data:

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126,
126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152,
160, 180

Number of values = 27

Choose 3 equal-frequency bins:

Each bin = $27 \div 3 = 9$ values

Bin 1:

110, 112, 115, 115, 118, 120, 120, 122, 124

Smoothed:

110, 110, 110, 110, 124, 124, 124, 124, 124

Bin 2:

124, 126, 126, 126, 126, 130, 135, 135, 138

Smoothed:

124, 124, 124, 124, 124, 138, 138, 138, 138

Bin 3:

138, 138, 140, 145, 150, 152, 160, 180

Smoothed:

138, 138, 138, 138, 138, 180, 180, 180, 180

Final Smoothed Data:

110, 110, 110, 110, 124, 124, 124, 124, 124,
124, 124, 124, 124, 124, 138, 138, 138, 138,
138, 138, 138, 138, 138, 180, 180, 180, 180

3) Equal-Width Binning for Risk Categorization

Given Data:

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2,
30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

$$\text{Minimum} = 18.5$$

$$\text{Maximum} = 39.5$$

$$\text{Range} = 39.5 - 18.5 = 21$$

$$\text{Number of Bins} = 3$$

$$\text{Bin Width} = 21 \div 3 = 7$$

Bins:

$$\text{Bin 1: } 18.5 - 25.5$$

$$\text{Bin 2: } 25.5 - 32.5$$

$$\text{Bin 3: } 32.5 - 39.5$$

Smoothed Data (Closest Bin Boundaries):

18.5, 18.5, 18.5, 25.5, 25.5, 25.5, 25.5, 25.5,
32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5,
39.5, 39.5

Answer:

$$\text{Range} = 21$$

$$\text{Bin Width} = 7$$

$$\text{Number of Bins} =$$

4) Data Reduction using Bin Medians

Given Data (Sorted):

120, 140, 145, 145, 148, 150, 150, 150, 155,
155, 155, 155, 160, 160, 165, 170, 170, 175,
175, 175, 175, 180, 190, 200, 210, 230, 300

Total Values = 27

Number of Bins = 3

Each Bin = 9 values

Bin 1:

120, 140, 145, 145, 148, 150, 150, 150, 155

Median = 148

Bin 2:

155, 155, 155, 160, 160, 165, 170, 170, 175

Median = 160

Bin 3:

175, 175, 175, 180, 190, 200, 210, 230, 300

Median = 190

Smoothed Data (Bin Medians):

148, 148, 148, 148, 148, 148, 148, 148, 148,
160, 160, 160, 160, 160, 160, 160, 160, 160,
190, 190, 190, 190, 190, 190, 190, 190, 190

Answer:

Number of Bins = 3

Median of Bin 1 = 148

Median of Bin 2 = 160

Median of Bin 3 = 190

5) Standardization of Patient Age Data

Given Data:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

$$\begin{aligned}\text{Mean } (\bar{X}) &= (18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + \\ &56 + 28) / 10 \\ &= 330 / 10 \\ &= 33\end{aligned}$$

(a) Mean Absolute Deviation (MAD)

$$x_i - \bar{X} = 15, 11, 8, 9, 5, 10, 0, 2, 23, 5$$

$$\begin{aligned} \text{MAD} &= (15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5) / 10 \\ &= 88 / 10 \\ &= 8.8 \end{aligned}$$

Standard Deviation (σ) ≈ 10.75

(b) Z-score for the First Four Observations

Formula:

$$z = (x - \bar{X}) / \sigma$$

$$\text{Age 18 : } z = (18 - 33) / 10.75 = -1.40$$

$$\text{Age 22 : } z = (22 - 33) / 10.75 = -1.02$$

$$\text{Age 25 : } z = (25 - 33) / 10.75 = -0.74$$

$$\text{Age 42 : } z = (42 - 33) / 10.75 = 0.84$$

Answer:

$$\text{Mean} = 33$$

$$\text{MAD} = 8.8$$

Z-scores:

$$18 \rightarrow -1.40$$

$$22 \rightarrow -1.02$$

$$25 \rightarrow -0.74$$